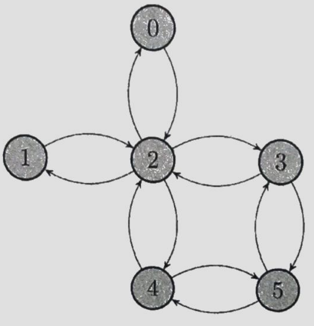
EE 595H : Stochastic Models

A. This is **NOT** an open book exam. The exam is for **40 Marks** and of duration **150 Minutes**. The marks for each question is indicated at the beginning of the question.

B. The question paper consists of **SEVEN** questions printed on **TWO** A4 sides. Answer to each question should begin on a **FRESH** side of the answer book.

Question 1. (6 marks)

Let $X_0, X_1, X_2, \ldots$ be a homogeneous Markov chain with state space $S = \{0, 1, 2, 3, 4, 5\}$. Suppose that the Markov chain evolves according to the transition graph depicted in the figure below, where from any state, we move to a neighbouring state uniformly at random.



(a) Compute the transition matrix of the Markov chain.

(b) Is the chain irreducible? Is it periodic or aperiodic?

(c) Find all stationary distributions of the Markov chain.

Question 2. (6 marks)

Describe in full detail the Box-Muller method. How can the method be used to obtain samples from the multivariate normal distribution with mean $\mu \in \mathbb{R}^2$ and variance $\Sigma \in \mathbb{R}^{2\times 2}$.

Question 3. (6 marks)

A $k \times k$ matrix $P = [P_{ij}]$ is said to be doubly stochastic if $P_{ij} \geq 0$ and $\sum_{j=1}^{k} P_{ij} = 1$ for all i and $\sum_{i=1}^{k} P_{ij} = 1$ for all j.

(a) Consider a homogeneous Markov chain with a doubly stochastic transition matrix Q. Show that uniform distribution is a stationary distribution for this Markov Chain.

(b) Consider a homogeneous Markov chain with a transition matrix R for which the uniform distribution is a stationary distribution. Show that R is doubly stochastic.

Question 4. (6 marks)

Suppose N balls are distributed among two boxes, A and B. At each move, a ball is selected uniformly at random and transferred from its current box to the other. Let X_k denote the number of balls in box A after k moves. Note that $X_0, X_1, X_2, \ldots$ form a homogeneous Markov chain.

(a) Describe the state space, the transition matrix and the transition diagram.

(b) Find all stationary distributions of the Markov chain.

(c) Is the Markov chain reversible? Check for detailed balance.

Question 5. (6 marks)

A student has N books on a bookshelf. All books are equally interesting for his research, so that when he decides to take one from his library, it is in fact chosen at random. Being lazy, he does not put the book back to where it was, but instead at the beginning of the collection, in front of the other books. The arrangement on the shelf can be represented by a permutation σ of $\{1, 2, \ldots, N\}$, and the evolution of the arrangement is therefore a homogeneous Markov chain on the set of permutations $\mathcal{S}_N$.

(a) Is the chain irreducible? Is it periodic or aperiodic?

(b) Find all stationary distributions of the Markov chain.

(c) Is the Markov chain reversible? Check for detailed balance.

Question 6. (4 marks)

We would like to use the Metropolis algorithm to get samples from the binomial distribution $Bin(n, p)$ with parameters $n = 5$ and $p = \frac{1}{2}$. Consider the simple random walk on the state space $S = \{0, 1, 2, \ldots, 5\}$ with transition probability $Q_{ij} = \frac{1}{2}$ for $|i - j| = 1$, $Q_{00} = \frac{1}{2}$ and $Q_{nn} = \frac{1}{2}$. Compute the transition matrix P obtained using the Metropolis algorithm.

Question 7. (6 marks)

On a space $S = \{1, 2, \ldots, k\}$, a target distribution $\pi = (\pi_1, \pi_2, \ldots, \pi_k)$ and an irreducible aperiodic homogeneous Markov chain with transition probabilities Q_{ij} are given. Assume $\pi_i > 0$ and $Q_{ij} > 0$ iff $Q_{ji} > 0$. Consider acceptance probabilities defined as

$$\beta_{ij} = \frac{\pi_j Q_{ji}}{\pi_j Q_{ji} + \pi_i Q_{ij}}$$

and a new chain with transition probabilities $P_{ij} = \beta_{ij} Q_{ij}$.

(a) Show that this new chain is irreducible and aperiodic.

(b) Show that this new chain has π as a reversible distribution.

END